

Estimating fuel load for wildfire risk assessment using earth observation data: A case study in Southwestern Australia

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Abstract

Earth observation (EO) data can support wildfire risk assessment by estimating fuel load, a critical factor of fire behaviour models. We use EO data to estimate fuel load at regional scales with a case study in southwest Western Australia from 2001 to 2021. Output maps derived from the EO product MCD64A1 (MODIS) are compared with those derived from a manually compiled fire history dataset (MCFH). Results suggest that EO data are effective in maintaining spatially and temporally complete records of fires, as MODIS reports 11,019 more hectares of burned areas associated with wildfires than MCFH over the study period. EO is more effective in mapping wildfires than prescribed burns, as the spatial overlapping ratio is higher for wildfires (0.63) than prescribed burns (0.42). When estimating fuel load, the high agreement (weighted kappa of 0.91) between MODIS and MCFH results from grassland covering the majority of the landscape in the study area. However, the agreement is reduced for other vegetation types — 0.24 for pine, 0.36 for mallee heath, 0.39 for shrubland, and 0.58 for forest. Results also indicate that satellite data has lower effectiveness in detecting small and under-canopy fires such as prescribed burns. This study highlights the potential for using EO data in the management of wildfire risk as the data are easily accessible and reliable, as well as efficient and cost-effective to use, and they provide the opportunity to perform wildfire risk assessments and develop risk reduction strategies at regional scales.

Keywords: MCD64A1, Fire history dataset, Burned area, Fuel accumulation, Wildfires, Prescribed burns.

1. Introduction

Wildfires are a worldwide phenomenon, causing significant losses of property, human life and ecosystem assets (Bowman et al., 2009; Moritz et al., 2014), as well as changes to hydrological regimes (Partington et al., 2022). These are likely to become worse as a result of climate change (Bowman et al., 2009; Canadell et al., 2021) and urban expansion (Chuvieco, Giglio, & Justice, 2008; Norman et al., 2021). Consequently, the ability to quantify wildfire risk and identify the most suitable areas for mitigation is becoming increasingly important. This requires accurate estimates of fuel load at regional scales, as such estimates are a key input to wildfire modelling approaches used to quantify and mitigate wildfire risk (Cruz et al., 2018; Gould et al., 2011; Keane, 2012).

The spatial and temporal distribution of fuel load is generally obtained with the aid of fuel accumulation models (Keane, 2012; Thonicke et al., 2001; Zazali et al., 2020), which represent the growth of vegetation and litter decomposing over time (McCarthy et al., 1999). These models require knowledge of the spatial and temporal distribution of fuel age, which is a function of the timing and extent of historical fuel disturbance events, such as wildfires, prescribed burns, mechanical removal of fuel and clear-fell logging (Farris et al., 2010; Johnson & Gutsell, 1994; Keane, 2012; Thonicke et al., 2001; Zazali et al., 2020).

Records of fuel disturbance events are commonly compiled manually from sources such as ground intelligence (e.g. using GPS), air intelligence, line-scan imagery, and maps translated from satellite images (AFAC, 2022). However, this approach is not well suited to support the modelling of fuel age and fuel load at regional scales, as it is often labour-intensive, time-consuming and costly (Farris et al., 2010). In addition, it can result in significant inaccuracies due to the inherent subjectivity associated with ground-based observation methods (e.g. field mapping) when records of fuel disturbance events are derived from ground intelligence (Chuvieco et al., 2019) and due to inaccuracies resulting from human interpretation associated with data pre-processing and transmission when records of fuel disturbance events are derived from satellite data (NAFI, 2021).

Inconsistent mapping criteria and data updating cycles also complicate the regional application of manually compiled historical fire events, as this often requires the sharing of data across different agencies and jurisdictions (Hehir et al., 2022). In addition, some historical fire events are recorded with the data collection date, rather than the actual date when the fires occurred

(Dixon et al., 2022), resulting in errors that can propagate with time when modelling fuel age and fuel load.

This study proposes an approach to address the limitations of the methods outlined above to estimate fuel load by replacing manually compiled fire history datasets with those derived from earth observation (EO). EO data are widely available and have systematic global coverage (Chuvieco et al., 2020), provide accurate and up-to-date information on areas affected by fires (Chuvieco et al., 2019), and provide low-cost and quantitative means to map burned areas (Szpakowski & Jensen, 2019). It should be noted that while remote sensing techniques such as LiDAR have been used to develop physically based models to estimate fuel load (Chen et al., 2017; Skowronski et al., 2007), thereby overcoming the need to map the extent of historical fire events, their application has only been tested on relatively small case study areas. In contrast, EO products designed to identify burned areas, which are by far the most significant type of fuel disturbance events (Canadell et al., 2021; Chuvieco et al., 2019; Giglio et al., 2018), have been applied successfully at regional, country and global scales (Belenguer-Plomer et al., 2019; Boer et al., 2008; Chuvieco, Englefield, et al., 2008; Chuvieco et al., 2016; Laurent et al., 2018; Miranda et al., 2022; Pulvirenti et al., 2023). In addition, the application of methods designed to estimate fuel structure is significantly more time and resource intensive, making them more difficult to utilise in practice.

Therefore, the aim of this paper is to demonstrate the utility of EO to overcome the challenges of manually compiled fire history data and illustrate the translation from EO to fuel load calculations with the aid of empirically based fuel accumulation models. This aim will be achieved through the following objectives: 1. Propose an approach that enables EO burned area (BA) products to be used to estimate fuel load at regional scales. 2. Evaluate the framework via a case study in a fire-prone landscape, southwest Western Australia, to better understand the level of agreement between BA maps, fuel age and fuel load maps derived from EO and a manually compiled fire history dataset; the effectiveness of EO in mapping wildfires and prescribed burns, the effectiveness of EO in BA mapping in remote regions, and the strengths of blending EO and manually compiled fire history datasets.

The remainder of this paper is organised as follows: Section 2 introduces the case study, materials and methods used in this study to estimate fuel load using EO. Section 3 demonstrates the results in relation to existing relevant research and discusses implications of this research. Section 4 concludes the study.

2. Materials and methods

2.1 Case study area

The study area is in southwest Western Australia (WA), roughly between 31°S~35°S and 115°E~120°E (Fig.1). WA is a significant bushfire risk area, with 90% of its landscape prone to bushfires (DFES, 2022). It has distinctly dry and hot summers (BOM, 2022), with the fire season spreading from October to March (Russell-Smith et al., 2007). Vegetation cover of the study area is dominated by eucalypt forest, woodland, grassland, and mallee heath (AFAC, 2022). For many years, government agencies in WA have used manually compiled fire history data to estimate fuel load and use the information to inform their fuel management policies (Hehir et al., 2022).

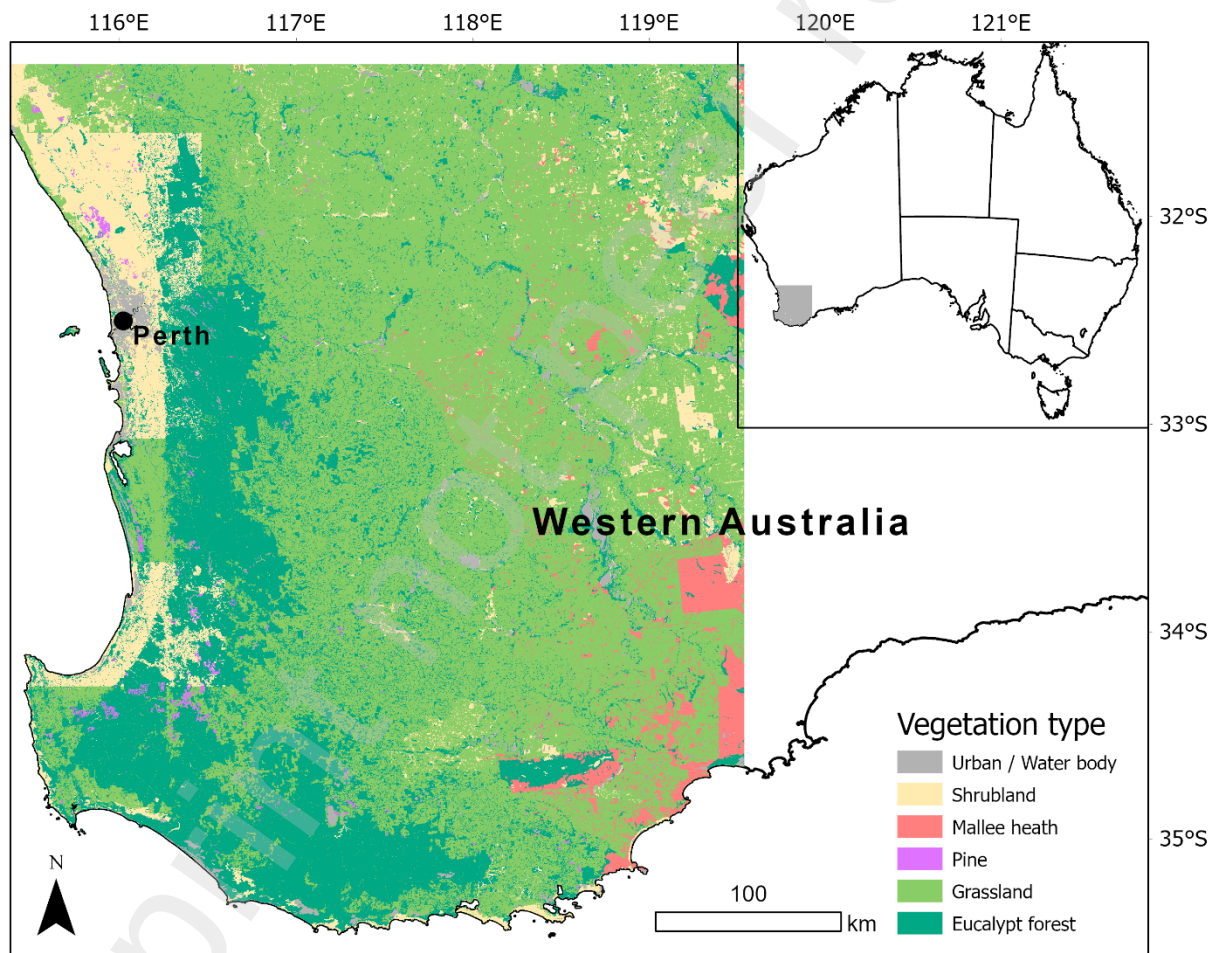


Fig.1. Map of the study area in southwest Western Australia with different vegetation cover types presented.

2.2 The approach of using earth observation data to estimate fuel age and fuel load

As depicted in Figure 2, the approach consists of five main steps. The first step is to obtain EO products that are capable of representing the fuel disturbance history for the region of interest.

EO data products that provide information on BA include MCD64 (Giglio et al., 2009), MCD45 (Roy et al., 2008), Geoland 2 (Tansey et al., 2012), VGT_cci, MERIS_cci (Chuvieco, 2013), AVHRR LTDR (Otón et al., 2021), MERGED_cci (Chuvieco, 2013), and Fire_cci (Chuvieco et al., 2018; Chuvieco et al., 2016). These products all have global coverage, are freely available and easily accessible through online platforms such as NASA AppEEARS and the European Space Agency Climate Change Initiative's open data portal (Padilla et al., 2015).

We select MCD64A1 Version 6 product (called MODIS hereafter) as an exemplar EO data for analysis. MODIS is a burn mask product at $500\text{ m} \times 500\text{ m}$ resolution (Giglio et al., 2020), which is generated through compositing the daily surface reflectance and daily active fire masks (Giglio et al., 2018; Vermote et al., 2002). This product is selected because it has a daily temporal resolution (Giglio et al., 2018), low occurrence of false positives (Hantson et al., 2013), high accuracy of BA mapping compared with other products such as MCD45, Geoland 2 and Fire_cci (Padilla et al., 2015; Vermote et al., 2002), and covers a temporal archive from 2000 to 2022 (Giglio et al., 2020). Although this product has been discontinued, the product and its algorithm remain the baseline for its replacements, such as the Visible Infrared Imaging Radiometer Suite (VIIRS) VNP64A1 BA product (Giglio et al., 2018). MCD64A1 contains information on the burn date, uncertainty score of the date, and quality assurance indicators (Giglio et al., 2015). Images covering the study area are downloaded in WGS84 projection from the USGS website (USGS, 2022), with a total BA of approximately 3.6 million hectares for the study period. The downloaded images are in *.hdf* format, each representing a composite of burn date in a calendar month.

— this truncation point is selected to align with the calculation of the fire behaviour index models in the Australian Fire Danger Rating System (Matthews, 2022).

The fourth step is to update the fuel age map using the BA maps. As can be seen in Step 4, new BA in a given year is depicted by a white colour which indicates a reset to a value of 0 for fuel age, while non-BA is depicted by a darker colour indicating that fuel age in a given year being one year older than that of the previous year. The maximum value for fuel age is 46 years, signifying areas not burned for the 46-year period from 1975 to 2021.

The fifth step is to update the map of fuel load (ton/hectare) maps using fuel age maps and a vegetation map. Fuel load refers to the amount of fuel that can be consumed if a fire occurs (Tolhurst & Kelly, 2003), which can be obtained using fuel accumulation models. These models can take on different forms, including single-component models (Olson, 1963), dual-component models (Bunnell & Tait, 1974), logistic growth models (Zazali et al., 2020), regression models (Chen et al., 2017), random forest models (McColl-Gausden et al., 2019), or artificial neural networks (D'Este et al., 2021). We use the Olson curve (Olson, 1963) as the fuel accumulation model, as it has been applied extensively to estimate fuel load in Australia (Gould et al., 2011; Matthews et al., 2012; McCarthy et al., 1999; Tolhurst & Kelly, 2003; Volkova et al., 2019). We note that the Olson curve has limitations as it is ineffective at capturing elevated fuel load and estimating fuel accumulation in complex ecosystems (Zazali et al., 2020). However, the curve is considered to be the most suitable approach here, given the data available. In addition, it is beyond the scope of this research to explore the best fuel accumulation model for use within the proposed framework.

As part of Step 5, a vegetation (fuel) types at each grid cell provided by the WA Department of Fire and Emergency Services (DFES), which we determine to be the most reliable data available for the study region, is used in the fuel accumulation model. When using the Olson curve, fuel load can be calculated as a function of fuel type and fuel age as follows:

$$x_{i,j}(t_{i,j}) = \frac{L_{i,j}}{k_{i,j}}(1 - e^{-k_{i,j}t_{i,j}}) \quad (1)$$

where $t_{i,j}$ is the fuel age at spatial grid cell i,j , $x_{i,j}(t_{i,j})$ is the cellular litter density at spatial grid cell i,j for fuel age $t_{i,j}$, $L_{i,j}$ is the fuel accession rate at spatial grid cell i,j , which is a function of the fuel (vegetation) type at spatial grid cell i,j , and $k_{i,j}$ is the fuel decomposition rate at spatial grid cell i,j , which is a function of the fuel (vegetation) type at spatial grid cell i,j . The above equation is based on the assumption that litter accumulates linearly with time and that the decomposition of litter eventually balances this accumulation, at which point the mass plateaus

to a steady state after a given accumulation period (Appendix 1). The equation is applied to each $100\text{ m} \times 100\text{ m}$ grid cell over the case study area, resulting in 21 annual fuel load maps.

2.3 Methods of assessing the strengths of EO in estimating fuel load

2.3.1 Comparison with the manually compiled fire history dataset

For the purpose of comparison, the manually compiled fire history dataset (MCFH) is used in this study. The dataset records fire events in polygons from 1937 to 2022 and contains information such as burn date, fire perimeter, and fuel disturbance types (e.g. prescribed burns, wildfires, mining rehabilitation, clear-fell logging, etc.). Fire events taking place from 2001–2021 are extracted for analyses, with a total BA of approximately 2.5 million hectares. In total, during the study period, MCFH records 2,999 wildfires (~780,694ha) and 4,437 prescribed burns (~1,718,365ha). The mean and median BA size is 260ha and 7ha for wildfires, and 387ha and 19ha for prescribed burns. Similarly, MCFH burn date attribute is used to subset fire events in a fire year to create an annual BA raster map at $100\text{m} \times 100\text{m}$, the same resolution as the MODIS BA maps. This process is repeated for each fire year in the archive, generating 21 BA maps. Using the same approach, MCFH BA maps are used to generate fuel age and fuel load maps, which are then referenced to GDA 1994 Albers.

The streams of difference maps are depicted in Figure 3, which are generated separately from wildfires and prescribed burns, as previous studies have indicated that EO is more likely to identify BA caused by wildfires than those caused by prescribed burns (Dixon et al., 2022). This separation is achieved by applying burn type masks to the maps of fuel age and fuel load. As MODIS data does not include burn type labels, the burn type masks are generated using the MCFH burn type labels.

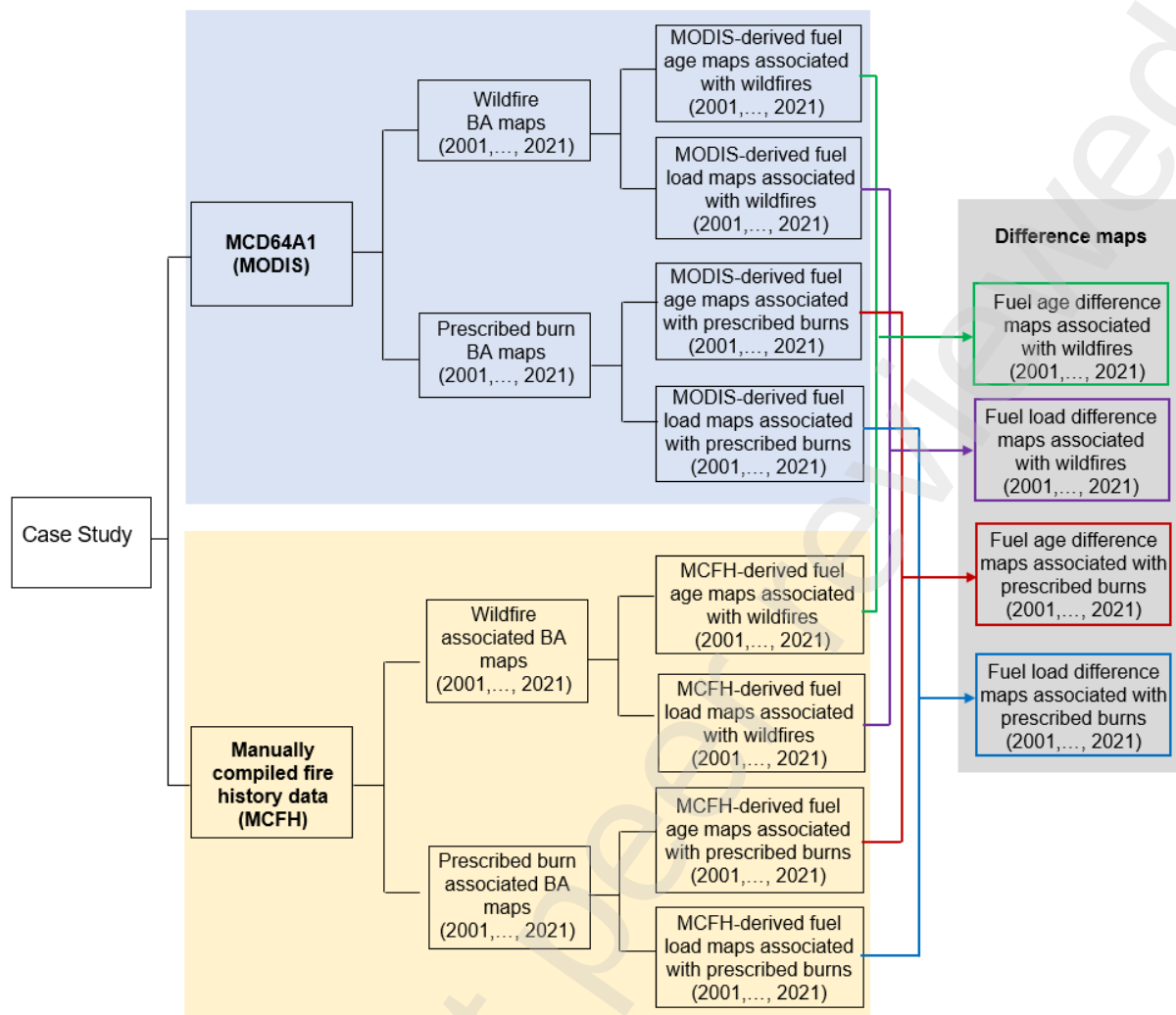


Fig. 3. Streams of difference maps derived from MODIS and MCFH for comparison.

2.3.2 Map comparison for overall agreement between MODIS and MCFH

BA maps derived from MODIS and MCFH are compared in three ways. Firstly, we count the number of MCFH events detected by MODIS, and divided this number by the total number of MCFH events in a given year to obtain the agreement ratio by fire events. It is worth noting that fire events are defined differently between the two datasets. MODIS does not include a label of fire events, as it records burned pixels instead. Obtaining fire events from MODIS involves spatial-temporal clustering of the burned pixels to generate fire patches, as discussed by Lizundia-Loiola et al. (2020) and Laurent et al. (2018). In contrast, MCFH clearly defines a polygon as a fire event. Secondly, the agreement is further examined by dividing the size of total overlapping BA by the size of the total MCFH BA. Lastly, we calculate the kappa score to assess the agreement of cell-by-cell reporting of BA. The value ranges from -1 (fully distinct)

to 1 (fully identical) (Cohen, 1968; Hagen-Zanker, 2006; Parente et al., 2023). Kappa is expressed by the equation:

$$K = \frac{p_o - p_c}{1 - p_c}$$

where p_o is the proportion of cells in agreement, namely, both MODIS and MCFH agree the cell as burned or unburned, and p_c is the proportion of agreement that is expected by random chance.

Fuel age and fuel load maps derived from MODIS and MCFH are compared using weighted kappa, which is given by the equation:

$$K_w = 1 - \frac{\sum w \times O_{ij}}{\sum w \times E_{ij}}$$

where w is the weighting factor, O_{ij} is the frequencies of observed agreement, and E_{ij} is the frequencies of expected agreement between the two datasets. The weighting method we use is “squared”. Weighted kappa is ideal for a cell-by-cell comparison between maps of ordinal values (Visser & De Nijs, 2006). To this end, we classify the fuel age value from continuous (1, 2, ..., 46) to an ordinal with 23 ranks (0-2, 3-4, ..., 45-46). Similarly, fuel load value is ordered into 16 ranks as 1-2, 3-4, ..., 31-32.

2.3.3 Spatial agreement between MODIS and MCFH

Spatial agreement is obtained by calculating overlapping extent for each individual MCFH event detected by MODIS. The corresponding equation is of the form:

$$\text{Overlapping extent} = \frac{BA_{MODIS}}{BA_{MCFH}}$$

where BA_{MODIS} is the size of BA mapped by MODIS for an overlapping fire event, which is obtained using the number of burned pixels multiplied by the spatial resolution, and BA_{MCFH} is the size of BA mapped by MCFH for the same fire event.

2.3.4 MODIS burned area mapping across the landscape

We assume there are different levels of effectiveness between MODIS and MCFH in mapping BA across the landscape. To investigate this assumption, a regional mask of the Accessibility/Remoteness index of Australia (ARIA+) is overlaid on BA maps derived from MODIS and MCFH. ARIA+ is a geographical classification system used to categorise and

measure the level of remoteness or distance from major population centres (Glover & Tennant, 2003). It divides Australia into five categories of remoteness with decreasing accessibility from *Major Cities*, *Inner Regional*, *Outer Regional*, *Remote*, to *Very Remote* (Appendix 2). This dataset is downloaded in shapefile format from the Australian Bureau of Statistics (ABS, 2022).

3. Results and discussion

3.1 Overall level of agreement

Figure 4 presents the overall level of agreement between MODIS and MCFH in deriving BA maps, fuel age and fuel load maps. In total, MODIS (36,307ha) reports more BA than MCFH (25,288ha) (Fig.4a). The average overlapping ratio between the two datasets is 0.13 by fire events and 0.40 by BA, with an average Kappa score of 0.33 (fair agreement) (Fig. 4b). Overall, there is an increasing trend in agreement over the 21-year period. This is likely to be caused by the integration of satellite images into fire history dataset in some regions, which has been applied in recent years (Hehir et al., 2022).

Figure 4c depicts the agreement between MODIS and MCFH in estimating fuel age and fuel load. The high weighted kappa in the initial years results from the identical baseline fuel age map, which produces a kappa score of 1. Overall, fuel age maps have a moderate agreement (0.45) which shows an increase from 2010, compared to a high agreement for fuel load maps with an average weighted kappa of 0.91. The high agreement for fuel load is moderated when investigating the kappa value through the lens of vegetation types. Fuel load for grassland shows full agreement throughout the study period, whereas for other vegetation types the agreement ranges from fair (0.24 for pine, 0.36 for mallee heath and 0.39 for shrubland) to moderate (0.58 for forest) (Fig. 4d). The fact that grassland covers the majority of the landscape in the study area drives the high agreement for the overall fuel load. This can be explained by the Olson curve in which fuel load for grassland recovers quickly to pre-disturbance levels

following a fire event (Cruz et al., 2015).

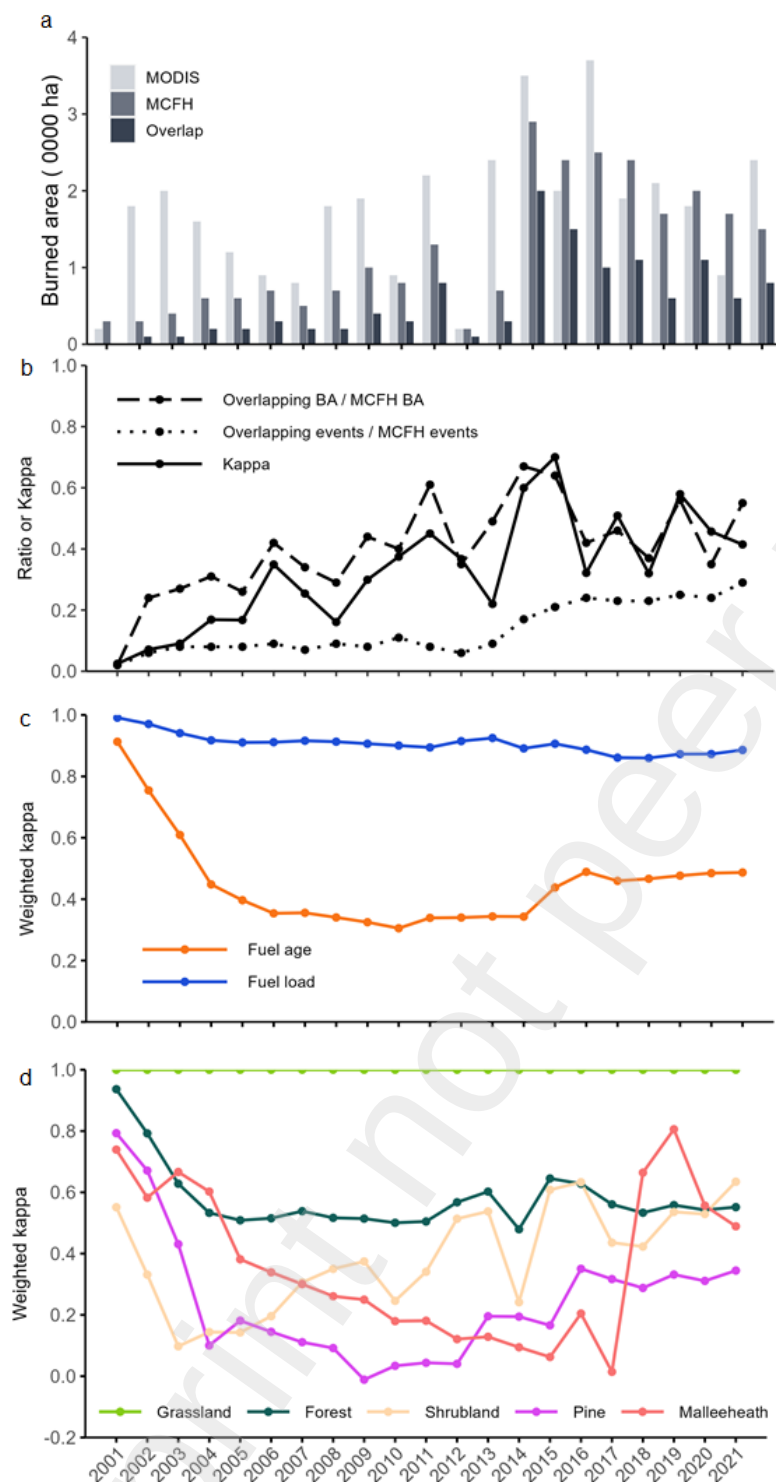


Fig.4. Overall level of agreement between MODIS and MCFH over the study period in (a) burned area mapping (maximum value is 40,000 ha), (b) overlaying ratio and kappa for burned area maps, (c) weighted kappa for overall fuel age and fuel load, and (d) weighted kappa for fuel load by vegetation types.

3.2 MODIS burned area mapping of wildfires and prescribed burns

Use of MODIS results in large areas where the estimated fuel age is substantially lower than when MCFH is used, as indicated by the blue patches in Figure 5a1. This difference in fuel age is caused by the identification of a larger number of areas that are burned by wildfires when the proposed framework is used, as this re-sets fuel age at the time of wildfire occurrence, thereby reducing the resulting fuel age. Given that MODIS has been demonstrated to have 98% of its detection of burned areas associated with actual fires (Hantson et al., 2013), these results indicate that significant areas that have burned as a result of wildfires have been missed by MCFH. This suggests that for the case study area, application of the proposed approach results in a significant increase in fuel age estimation accuracy in regions where fuel has been disturbed by wildfires.

The differences of fuel load between MODIS and MCFH are mostly concentrated in non-grassland ecosystems along the coastal regions whereas inland areas present minimal difference, as indicated by the blue patches in Figure 5a2. This is likely due to the nature of the fuel accumulation model used (i.e. the Olson curve). As can be seen by comparing Figures 5a1, 5a2 and the second-to-left panel of Figure 6, the regions where differences in fuel age are less likely to translate to differences in fuel load correspond to inland regions. These regions are mostly covered by grassland, which recovers quickly to pre-disturbance levels post wildfire events when simulated using Olson curve (Cruz et al., 2015).

In contrast to the trend for wildfire BA mapping, MODIS misses the majority of the prescribed burns, thereby continuing to increase fuel age where it should have been re-set to zero. As can be seen by comparing Figures 2 and 5b1, prescribed burns are conducted in areas of forested vegetation cover, as forests present a higher fuel hazard compared with that of other vegetation types (Burrows & McCaw, 2013; Collins et al., 2023). This produces significant differences in fuel load maps, where MODIS-derived fuel load is much higher than the MCFH-derived fuel load as MODIS fails to detect the majority of prescribed burns (Fig. 5b2).

As shown in the leftmost panel in Figure 6, the differences in fuel age obtained using MODIS and MCFH are relatively minor near the beginning of the simulation period in 2002, which is due to the use of the same baseline map of fuel age for both approaches. However, over the extent of the 21-year simulation, the impact of the relatively small number of wildfire events that are missed on an annual basis by MCFH is amplified, as they have a cumulative impact.

While it is not possible to determine specific reasons for MCFH's lack of identification of a large number of wildfire-caused burned areas, potential reasons include the error prone nature of using manual methods for BA mapping, as discussed in the Introduction. In addition, MCFH data is generally only obtained for land that is managed by public agencies, making it highly likely that any burns on private land are not recorded. This is where the universal coverage of EO has a significant advantage.

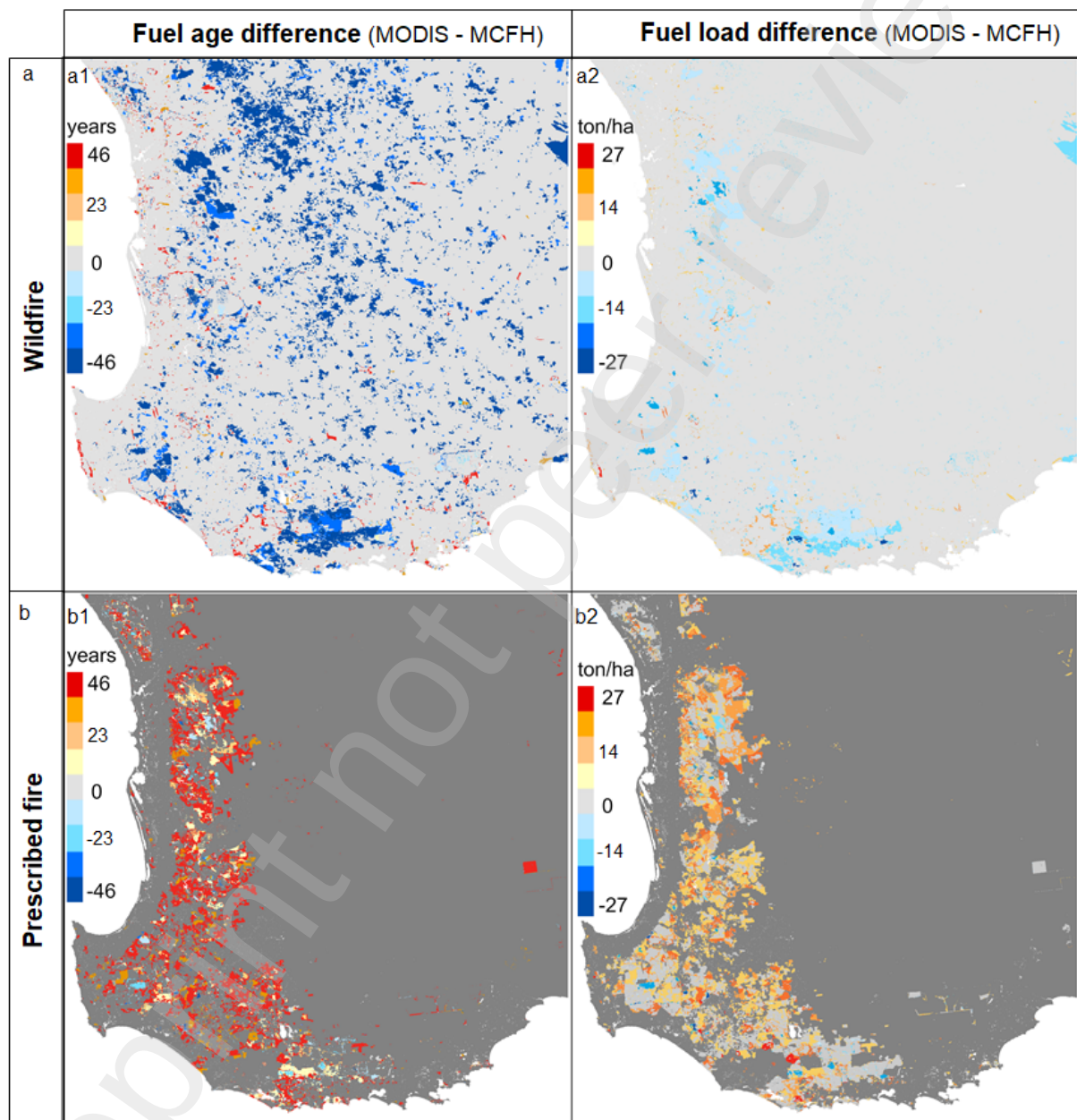


Fig. 5. Accumulated difference in fuel age and fuel load in 2021 obtained using MODIS and MCFH for burned areas associated with (a) wildfires and (b) prescribed burns. The equation of subtracting MCFH from MODIS applies to all maps. In a1 and b1, the fuel age difference value ranges from -46 to 46 years. Red denotes BA solely recorded in MCFH, light grey denotes no difference, and blue denotes the accumulation of BA solely mapped by MODIS.

In a2 and b2, fuel load difference value ranges from -27 to 27 ton/ha. Red denotes that the MODIS-derived fuel load is more than MCFH-derived fuel load, light grey denotes no difference, and blue denotes that MODIS-derived fuel load is less than MCFH-derived fuel load. The dark grey colour in the bottom row denotes null data as prescribed burns are not observed in these regions.

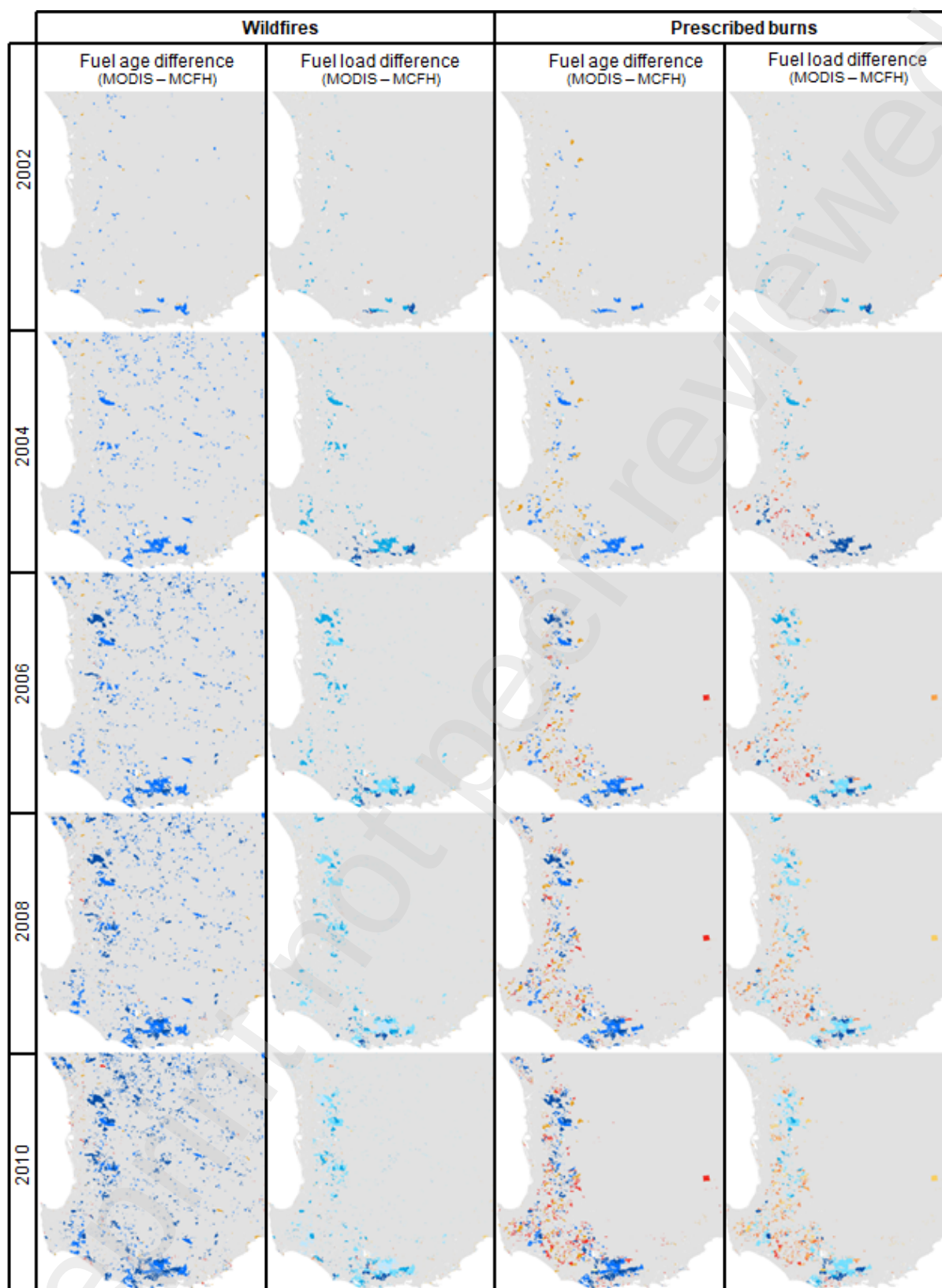
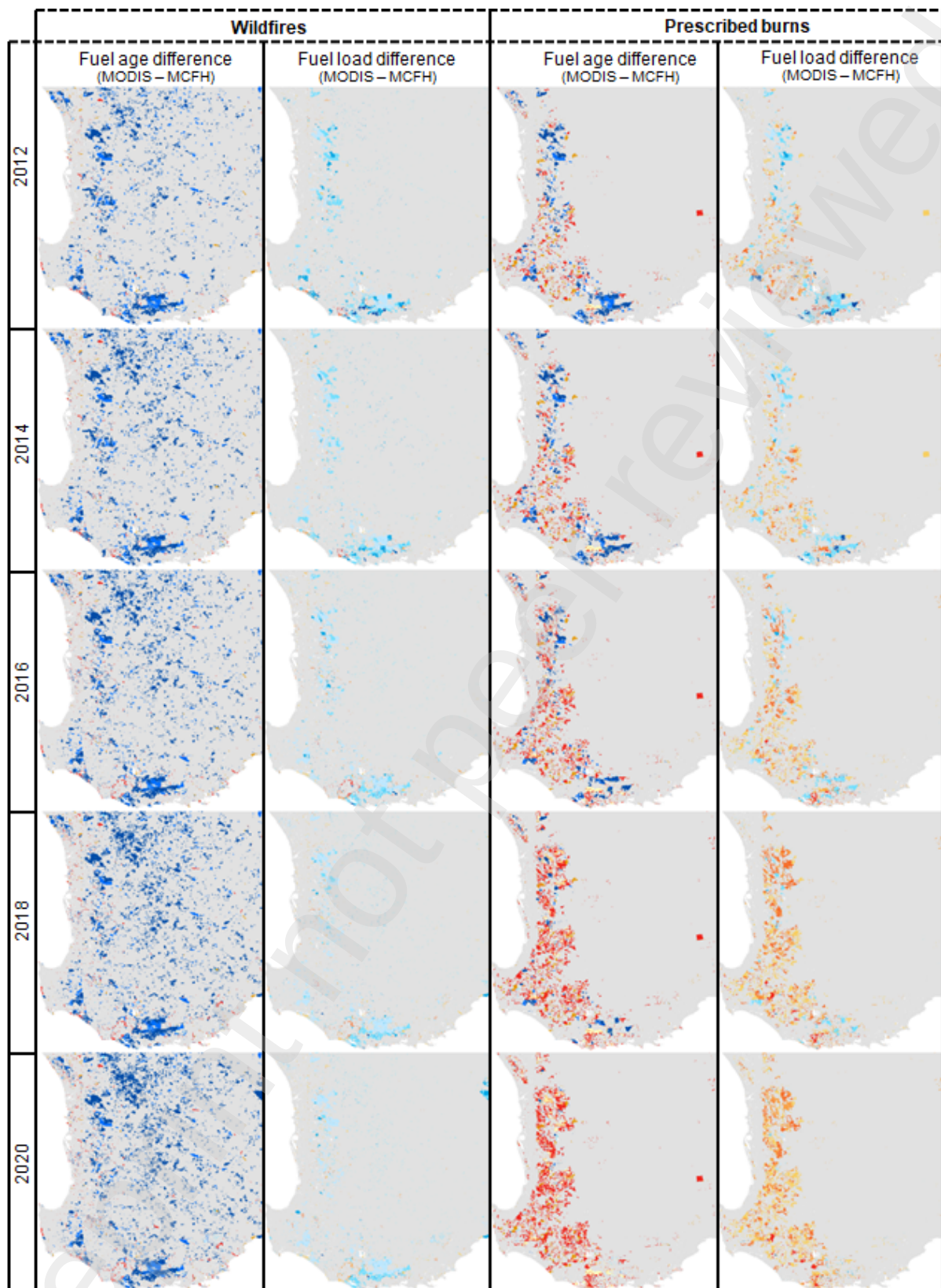


Fig. 6. Temporal evolution of differences in fuel age and fuel load from 2002 to 2020 at 2-year intervals partitioned based on fuel disturbance events caused by either wildfires or prescribed burns. The increasing colour saturation implies an increase in the magnitude of

differences. The equation of subtracting MODIS from MCFH applies to all maps. Blue denotes that fuel age and fuel load obtained using MODIS data are less than those derived from MCFH data, light grey denotes no difference, and red denotes that fuel age and fuel load obtained using MODIS data are greater than those derived from MCFH data.

Fig. 6 *continued.*



3.3 Spatial agreement between MODIS and MCFH

The result of the overlapping extent suggests that MODIS is more effective in mapping wildfires than prescribed burns (Fig. 7). The average overlapping extent is higher for wildfires

(0.63) than prescribed burns (0.42). It is well documented that prescribed burns are normally associated with lower intensity fires (Burrows & McCaw, 2013) whereas wildfires commonly have higher energy output and are more destructive to the vegetation, which increases the probability of a MODIS detection. Furthermore, as wildfires tend to have higher severity (Keeley, 2009) than prescribed burns — for example, wildfires are likely to result in complete canopy consumption (Dixon et al., 2022) — in such situations, BA associated with wildfires might be more visible to MODIS.

The spatial agreement analysis also shows that MODIS captures the extent of wildfires with higher precision. As shown in Figure 7a2, the black outline is the MCFH fire perimeter and the blue patches are the MODIS burned pixels. When overlaying the MCFH wildfire polygons with true colour images, we find that the fire perimeter overlaps with trail paths and road surrounding the BA. This is related to the data capture method employed in MCFH when using ground intelligence. For example, GPS mapping and ground observation by crews requires walking and driving access around the BA (AFAC, 2022, p11), which results in the overlap between the wildfire polygons and trail paths. Therefore, in some cases, manually compiled fire history datasets tend to draw the fire scar polygons beyond the actual fire perimeter, which leads to over-reporting of BA when mapping wildfire extent (Fig. 7 a2). This over-reporting is concerning, as it may result in underestimating fuel load in the unburned area, and hence the underestimation of wildfire risk.

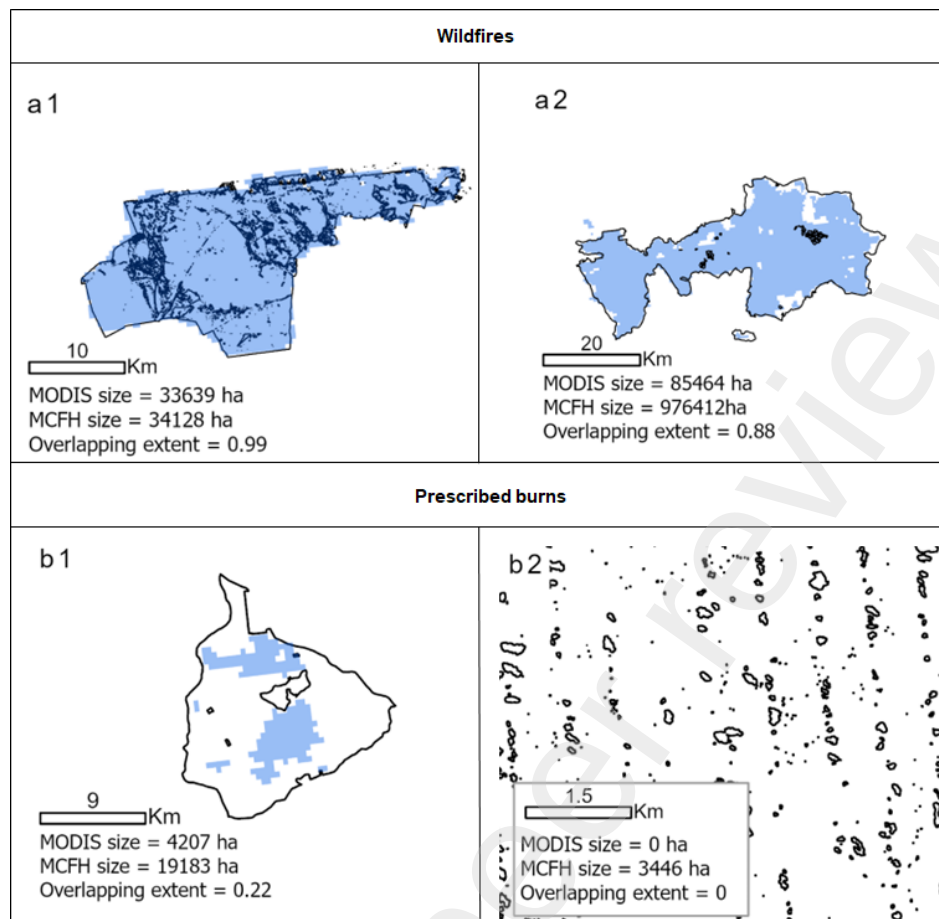


Fig. 7. Examples of overlap between extent of burned areas derived from MODIS data (blue shading) and MCFD data (outlines) for both wildfires and prescribed burns.

MODIS tends to be only capable of partially capturing the fire extent of prescribed burns (Fig. 7 b1). Most often, the captured fractions consist of areas that experienced higher intensity burns and subsequently correspond to a higher contrast of surface change in the imagery. Planning prescribed burns involves pre-mapping the fire extent and deciding the intensity (Burrows & McCaw, 2013). In practice, these burns are likely to produce uneven severity within the fire extent, resulting in the fact that only the part with high intensity can be captured by MODIS.

Another potential reason for missed or incomplete mapping of prescribed burns is due to the fragmentation of this burn type (Fig. 7 b2). They comprise sparse burn patches to suit specific fuel treatment needs (Penman et al., 2011). These patches can be very small and their detection by MODIS can be reduced given the coarse resolution of the satellite sensor. For example, if a burn patch is as small as $100\text{m} \times 100\text{m}$, it is unlikely to be effectively captured by MODIS as its resolution is $500\text{m} \times 500\text{m}$. This result aligns with the findings described by Roy and Boschetti (2009), Hawbaker et al. (2008), Hantson et al. (2013), Smith et al. (2022) and Giglio

et al. (2018), who discussed the limitations of MODIS in detecting small and highly fragmented fires.

Forest canopies also impact MODIS effectiveness in mapping prescribed burns. As discussed previously, the areas where prescribed burns are most frequently practiced correspond to forested areas, as forests present a higher fuel hazard compared with that of other vegetation types (Burrows & McCaw, 2013; Collins et al., 2023). Under such circumstances, prescribed burns are constrained to sub-canopy fuel layers (Canadell et al., 2021). In the presence of a dense canopy, the BA may be invisible, reducing the effectiveness of MODIS in capturing the fire scars (Fig. 8). In addition, prescribed burns often have an energy output below the minimum detection capabilities of MODIS. Consequently, this poses a challenge for mapping prescribed burns using MODIS, as discussed by Hawbaker et al. (2008), Vermote et al. (2002) and Dixon et al (2021).

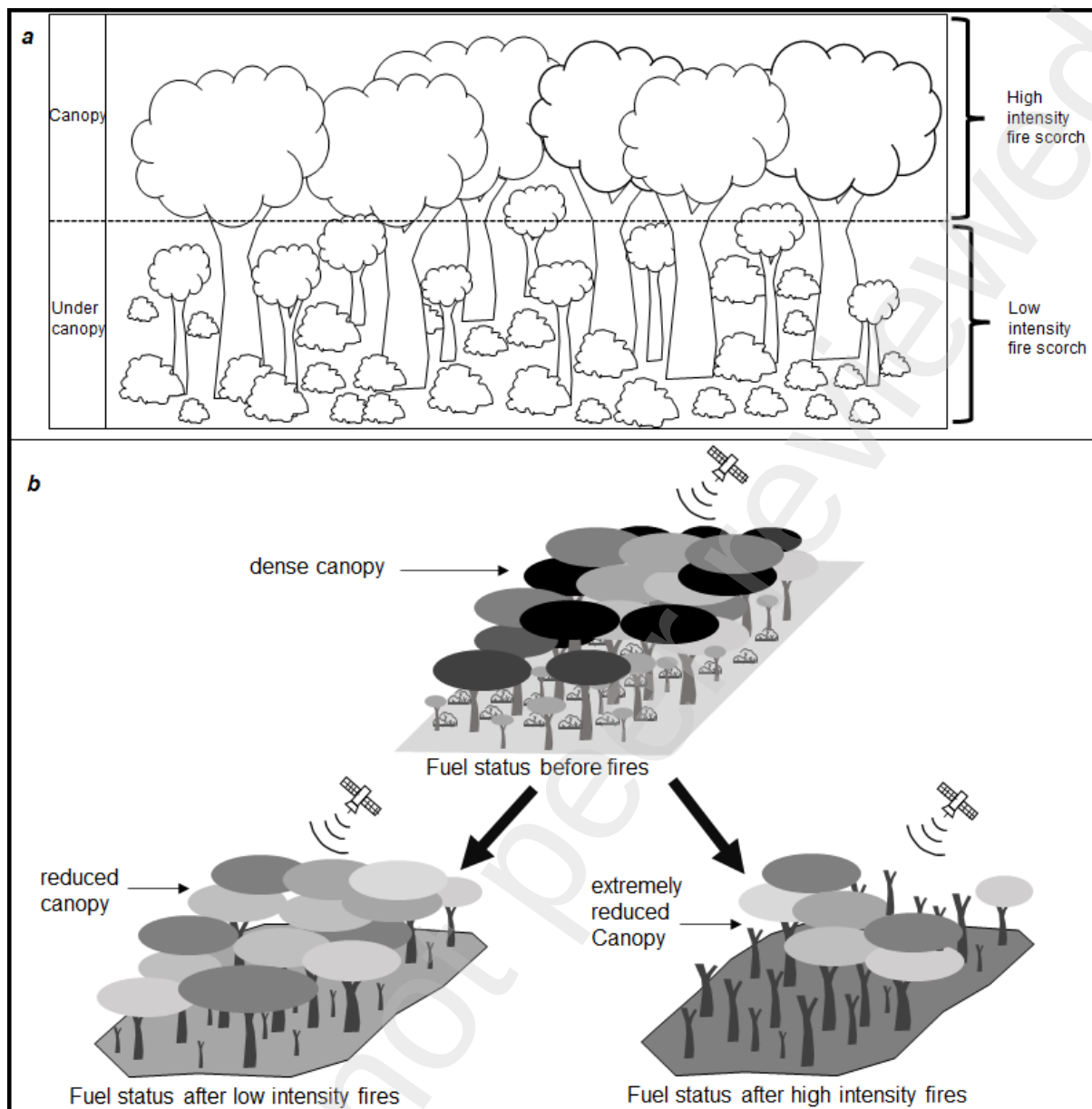


Fig.8. An illustration of (a) elevated fuel layers in forest and (b) how MODIS fails to map under-canopy burned areas. (Source: adapted from Gould et al., 2011; Dixon et al., 2022; and (Gale & Cary, 2022).

3.4 MODIS burned area mapping in remote regions

MODIS is effective in maintaining spatially and temporally complete records of BA. Figure 9 describes the results of overlaying ARIA+ with BA mapped by MODIS and MCFH. In *Major Cities* and *Inner Regional* regions, MCFH reports more BA than MODIS. MCFH focusses on these regions for data collection as these areas are populated with more assets and are easier to access compared to remote regions (personal communication with practitioners in WA, July 5, 2023). In contrast, in *Outer Regional*, *Remote* and *Very Remote* regions, MODIS records more BA than MCFH. As MODIS has been demonstrated to have low commission error (Hantson

et al., 2013; Padilla et al., 2015; Vermote et al., 2002), it is likely that the MODIS-mapped BA in remote regions are associated with actual fires, indicating MODIS' capability in capturing fires and BA across the landscape. This finding is in line with discussions on the strength of the MODIS algorithm which accounts for both surface reflectance change and thermal anomaly, providing the ability to capture BA with more precision (Giglio et al., 2018; Padilla et al., 2015). The algorithm enables the bias-free selection of the mapping areas, irrespective of the geographic, demographic or economic characteristics in the regions.



Fig.9. Burned area mapping in different remoteness regions by MODIS and MCFH from 2001 to 2021. The scale on the y-axis is different for each remoteness category. MCFH records more BA in less remote areas as prescribed burns are practiced in these regions.

MODIS records more BA in remote regions with its semi-automatic algorithm.

3.5 Merging MODIS and MCFH

Due to the nature of being manually compiled, MCFH is unable to guarantee that all fire-affected areas are mapped. As a result, estimating fuel age and fuel load solely based on MCFH is likely to be insufficient. However, MODIS can be used to integrate into MCFH to generate comprehensive BA maps. Figure 10 shows the difference maps of fuel age and fuel load derived from BA maps with and without integrating MODIS into MCFH. The maps indicate that MODIS results in significant differences in estimating fuel age as it records more areas with lower fuel age (i.e. more BA) and these areas are often in less urbanised regions. In terms of estimating fuel load, although the difference is not as significant as that for fuel age, the difference still spans across the landscape and these differences can be critical at the local scale, which can be used to inform practitioners' decisions on fuel management.

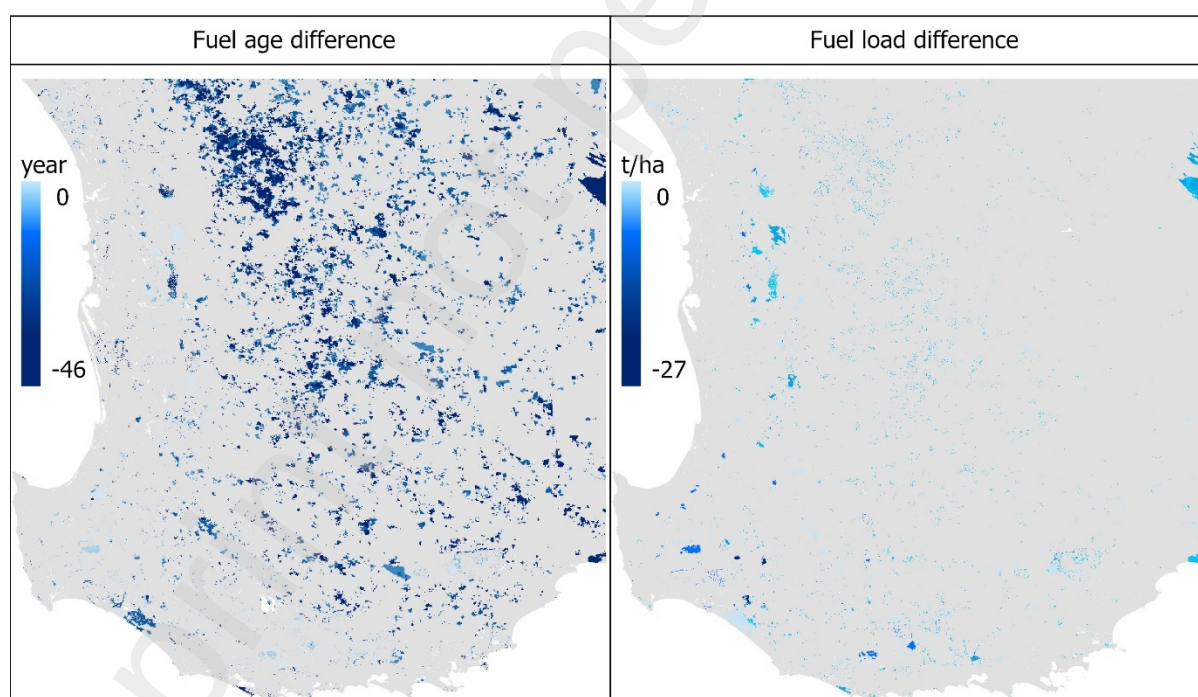


Fig.10. Fuel age and fuel load difference maps obtained by subtracting MODIS-and-MCFH maps from MCFH-only maps. Spatially, MODIS records more burned areas, hence more areas with younger fuel age and less fuel load, and correspondingly the negative value after the subtracting. Grey colour denotes no difference.

Whilst blending manual and satellite data can produce comprehensive BA maps, it is possible that EO products can be used as a replacement of manual data in the following circumstances:

- (i) where fire history data need to be updated frequently. For example, the semi-automatic algorithm of EO spares the need to deploy ground crews for field mapping, which saves a considerable monetary investment in human resources and equipment.
- (ii) where capability is limited. For example, countries lacking the capability to collect fire history data can utilise EO data to generate fire history datasets and use the information to support their fuel management decisions.
- (iii) where fuel hazard at regional scales needs to be mapped rapidly and/or remotely. For example, EO data provides near-real-time information on new BA associated with wildfires. As such, it can be used to support fire managers' understanding of fire extent shortly after the occurrence of such events. This timeliness is critical to supporting instant fuel and fire management decisions (Birchall et al., 2022, p20-25).

4. Conclusions

The quantification and management of wildfire risk is becoming increasingly important around the world. Modelling wildfire risk is underpinned by estimates of the spatio-temporal evolution of fuel load. This requires estimates of fuel age, which are affected significantly by events that disturb fuel, such as wildfires and prescribed burns. Consequently, estimates of the spatio-temporal distribution of these disturbances at regional scales is critical to the quantification and management of wildfire risk.

In this paper, an approach is introduced for estimating the spatio-temporal distribution of fuel load at regional scales using earth observation data. This increases the efficiency and cost-effectiveness of obtaining estimates of burned areas, and hence fuel load. The approach is flexible and easy to implement, as it enables users to select which earth observation data that are most suited to their application and fits in with existing approaches of estimating fuel loads.

The utility of the proposed approach is assessed on a regional-scale case study of a fire-prone area in south-west Western Australia. Results indicate that the use of EO is able to generate more accurate estimates of the spatio-temporal distribution of fuel age and fuel load for fuel disturbance events caused by wildfires. This is likely to be due to a combination of the increased efficiency and cost-effectiveness of the proposed approach. In contrast, EO is not able to identify the majority of fuel disturbance events caused by prescribed burns. This is likely to be due to a combination of the low intensity and fragmented nature of prescribed

burns, as well as their lack of disturbance of the upper canopy of vegetation, making it difficult for EO data to detect them. This limitation, however, can be complemented by merging EO data and manually compiled fire history data as prescribed burns are pre-recorded in the latter.

The results from this study highlight the potential for using EO data in the management of wildfire risk. As EO data are easily accessible and reliable, as well as efficient and cost-effective to use, they provide the opportunity to perform wildfire risk assessments and develop risk reduction strategies at regional scales, especially in remote areas or where resources are limited. As demonstrated in this paper, even in well-resourced areas where manually compiled fire history data is available, the use of EO data can result in the identification of a larger number of fuel disturbance events caused by wildfires, enabling a more accurate assessment of bushfire risk and risk reduction strategies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

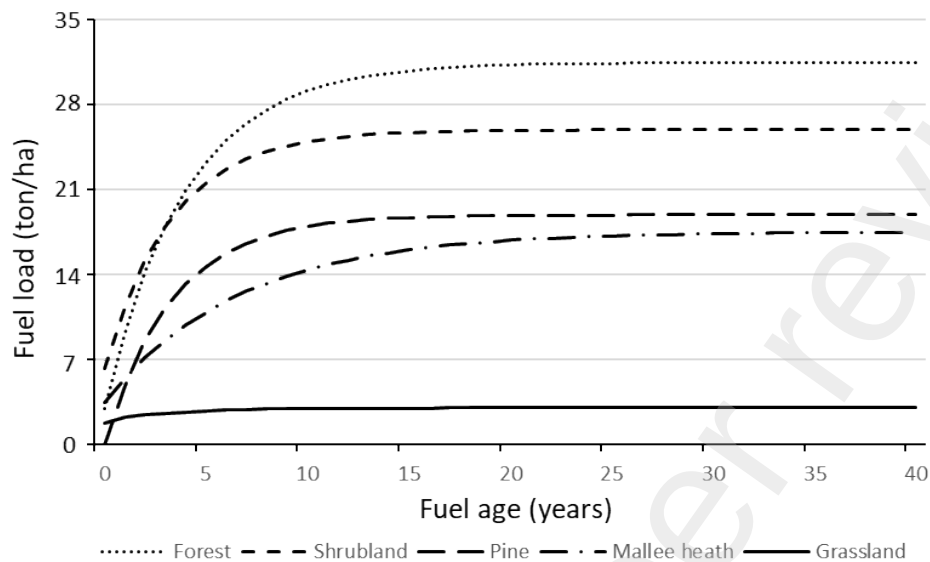
The Earth Observation dataset presented in this study are available on NASA AppEEARS. The script for processing datasets and calculating fuel age and fuel load is available on request to the corresponding author.

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Appendices

Appendix 1: An illustration of fuel accumulation trend for different vegetation types estimated using Olson curve based on randomly generated value, in which the fuel accumulation for grassland shows a plateau trend.



Appendix 2: Accessibility/Remoteness index of Australia (ARIA+) in the study area (Source: Australian Bureau of Statistics, 2022)



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